Kuber: Geometry-General Neural Surrogates for Conjugate Heat Transfer

Shubh Jain
Kuber.ai
github.com/ShubhJain007/Kuber

Abstract*—Conjugate heat transfer (CHT) — the coupled transport of heat through a solid conductor and a surrounding moving fluid — governs the thermal design of heatsinks, cold plates, heat exchangers, power electronics, and battery packs. High-fidelity CFD resolves these fields but costs minutes to hours per geometry, prohibitive inside a design loop. We present Kuber, an open framework for building neural surrogates of CHT, and SurfaceGeoTransolver, a geometry-general model that predicts the full steady field *(U, T, p)* at an arbitrary query point cloud in a sub-second. The model couples a physics-attention transformer core with a surface-geometry encoder, so it ingests raw boundary geometry (a surface point cloud with normals) and requires no analytic signed-distance field, and it is conditioned on the full set of physical scalars plus a device-class flag. On the public SIMSHIFT heatsink benchmark it attains 12.14 K temperature RMSE on the medium out-of-distribution split, improving on the best previously published result (12.41 K) while using no unsupervised domain adaptation, which every reported baseline relies on. A single set of weights spans two physically distinct device classes — buoyancy-driven heatsinks in air and forced-convection liquid cold plates — and pretraining on a self-generated OpenFOAM corpus lowers out-of-distribution error further. We additionally verify numerical soundness: the predicted temperature gradient never exceeds the physical gradient at fin tips and corners (explosion fraction zero, no NaN/Inf). All results are reproducible with the released code and data.

***Index Terms*—conjugate heat transfer, neural operators, surrogate modeling, transformers, physics-informed machine learning, out-of-distribution generalization.**

I. Introduction

Thermal design is an inner loop. An engineer proposes a geometry — a fin array, a cold-plate channel layout — and must know the resulting temperature and flow field before iterating. The field is produced by solving the coupled Navier–Stokes and energy equations across a solid and a fluid region: *conjugate heat transfer* (CHT). Finite-volume CFD solves this accurately but slowly; in our measurements a single steady case takes a median of 22 minutes and up to nearly two hours for fine geometries. Sweeping thousands of designs, or closing an optimization loop, is therefore dominated by solver cost.

Neural surrogates promise to amortize this cost: learn the geometry-to-field map once, then evaluate it in milliseconds. Recent operator-learning and transformer methods have made this practical for external aerodynamics, but electronics-cooling CHT has received far less attention, has no widely used open benchmark harness, and no shared, license-clean data-generation recipe. Two requirements make the problem hard. First, *geometry generality*: a useful surrogate must accept arbitrary CAD, which rules out geometry encodings that presuppose an analytic shape. Second, *generalization*: the surrogate must predict geometries it never saw in training.

We make three contributions. (i) SurfaceGeoTransolver, a surface-conditioned physics transformer that predicts the full five-channel CHT field at any query point from raw boundary geometry and physical conditioning, reaching state of the art on a public benchmark without domain adaptation. (ii) A reproducible data engine generating a self-labeled OpenFOAM CHT corpus across fluids, regimes, shapes, and two device classes, with a convergence gate and a verified mesh-convergence protocol. (iii) An honest evaluation harness — per-field normalized error, near-wall fidelity, a numerical no-explosion proof, and a value-of-data ablation — with released code, data, machine-readable results, and an interactive viewer.

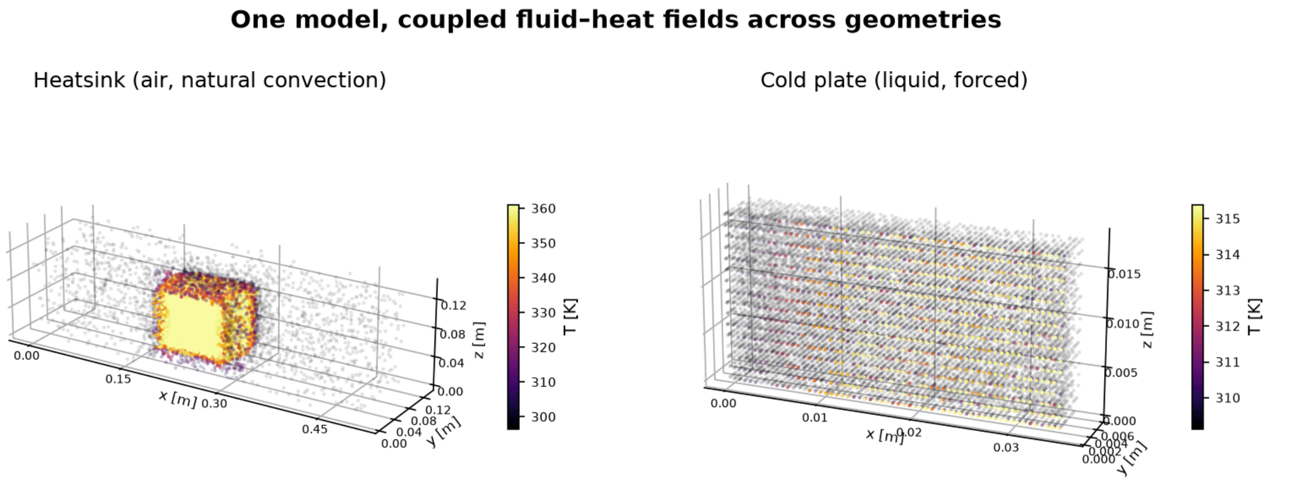

Fig. 1. Coupled fluid–heat fields predicted by a single model across two device classes: a heatsink in natural-convection air (left) and a liquid-cooled cold plate (right), colored by temperature (OpenFOAM ground truth shown).

II. Related Work

Neural operators. DeepONet [7] and the Fourier Neural Operator [6] established learning maps between function spaces; GINO [8] extended operator learning to large 3D geometries. These are strong on fixed domains but do not natively consume arbitrary boundary geometry with per-node queries. *Transformer PDE solvers.* Transolver [3] introduced physics attention over learned slices; UPT [4] and the anchored/branched variant (AB-UPT) [5] scale transformer surrogates to industrial CFD. Our core builds on the GeoTransolver implementation in NVIDIA PhysicsNeMo [10]; our contribution is the surface-conditioning branch and the CHT-specific conditioning and training recipe. *Geometry representations.* Signed-distance fields are data-efficient when an analytic shape exists, but none exists for arbitrary CAD; following the surface branch of AB-UPT [5] we encode the boundary directly as a point cloud with normals. *Spectral fidelity.* One-shot regression is spectrally biased; PDE-Refiner [9] restores high-frequency content, which we adopt as an optional head. *Benchmarks.* SIMSHIFT [2] is a public benchmark for distribution shift in engineering simulation; we use its heatsink task as our primary evaluation.

III. Problem Formulation

A CHT case consists of a solid conductor with boundary Γ immersed in a fluid domain Ω. Given the boundary geometry and operating conditions, we seek the steady field at any query point $x \in \Omega$:

$$f_\theta : (x, \Gamma, c) \mapsto (U_x, U_y, U_z, T, p_{\mathrm{rgh}}), \tag{1}$$

where U is velocity, T temperature, and $p_{\mathrm{rgh}} = p - \rho g h$ the reduced pressure. The conditioning vector c collects fluid properties (Prandtl number, density, specific heat, dynamic viscosity), boundary conditions (a wall temperature for heatsinks or a heat flux for cold plates, and the ambient/inlet temperature), inlet velocity, and a binary device-class flag. The boundary Γ is presented as surface points with outward unit normals. Nothing in the input assumes a parametric shape family.

IV. Method

A. SurfaceGeoTransolver

A *surface encoder* maps the boundary point cloud with normals to permutation-invariant geometry tokens via a shallow self-attention stack. A *local cross-attention* module produces, for every query node, a geometry descriptor by attending over its $k = 16$ nearest surface tokens, so each query carries a local view of the nearby boundary. This descriptor is concatenated with the broadcast conditioning vector to form the per-node embedding. The *core* is a GeoTransolver physics-attention transformer (256 hidden units, 12 layers, 8 heads, 64 physics slices, with multiscale local ball-query features) that regresses the five z-normalized targets. The model has approximately 14.3 M parameters. An optional PDE-Refiner [9] head can replace one-shot regression; results below use the one-shot model. Geometry may alternatively be supplied as a (directional) signed-distance field for parametric families, but the surface-encoder configuration is the one we report because it is the only one applicable to arbitrary CAD.

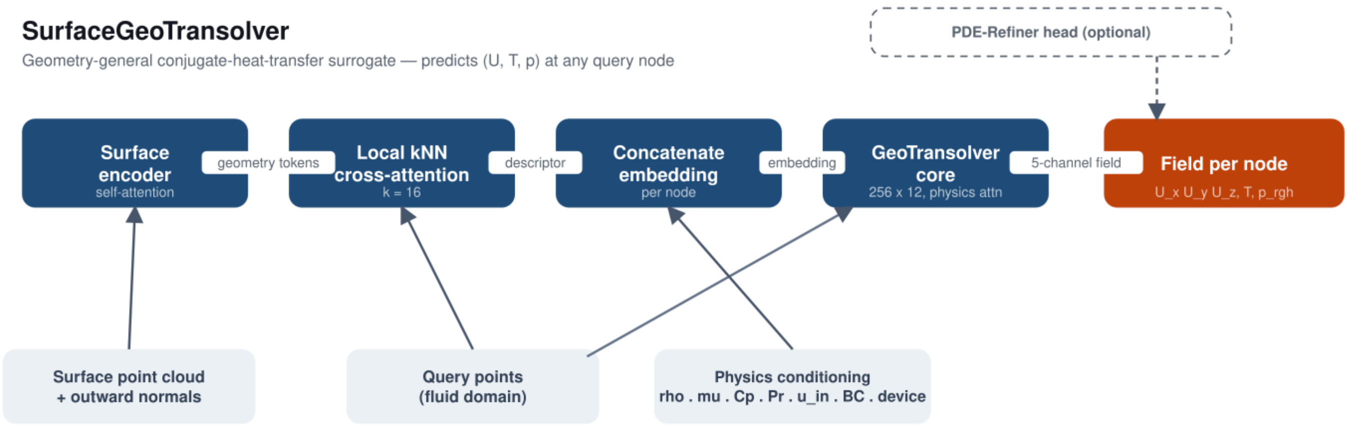

Fig. 2. SurfaceGeoTransolver. Three inputs — the boundary surface point cloud with normals, the query points, and the physics conditioning — feed a neural spine (surface encoder → local kNN cross-attention → concatenated per-node embedding → GeoTransolver physics-attention core) that predicts the five-channel field at every query node.

B. *Data Engine*

We generate CHT data with OpenFOAM's steady buoyant solver (buoyantSimpleFoam) over a single fluid region, treating the solid as a heated wall (heatsinks) or heat-flux surface (cold plates). A parametric generator samples geometry and conditions; the geometry is meshed with blockMesh and snappyHexMesh plus prism layers, solved, and exported to 16,384 fluid nodes with the five field channels, the boundary surface cloud with normals, and a JSON of conditions. The pipeline is resumable and gated: a case is written only after a converged solve, and a physics filter rejects unphysical results. Sampling is seeded, so the corpus regenerates deterministically. It is entirely self-generated (Table I).

TABLE I
DATA-ENGINE COVERAGE OF THE SELF-GENERATED CORPUS

Axis	Coverage
Fluids	air, water, oil (Pr≈292), glycol
Regimes	natural + forced convection
Shapes	fins, plate, cube, pin-fin; duct channels
Devices	heatsink (wall-*T*), cold plate (heat-flux)
Fidelity	≈1.4 M cells/case → 16,384 nodes

V. EXPERIMENTAL SETUP

Benchmark. Our primary, externally comparable evaluation is the SIMSHIFT [2] heatsink task at *medium* difficulty: train on fin counts 5–8, test on the shifted target of fin counts 10–12, which never appear in training. We train from scratch on the 222 training cases; normalizers are fit on the source training split only. *Metrics.* We report per-field RMSE in physical units and normalized RMSE (nRMSE); the benchmark's primary selector is mean nRMSE across the five fields. We additionally report near-wall *T*-RMSE and edge gradient ratios (Sec. VI-D). No baseline uses unsupervised domain adaptation (UDA) in our runs; the published baselines do, which is their best case. *Training.* Adam [11] with a warmup–stable–decay schedule, z-scored targets, per-case normalized frame. Baseline numbers are quoted from the SIMSHIFT paper.

VI. RESULTS

A. *SIMSHIFT Heatsink Benchmark*

On temperature — the design-critical field — SurfaceGeoTransolver improves on the best previously published result while using no UDA (Table II). The paper reports only temperature and velocity RMSE on the target domain; baseline parameter counts are not published but their released configurations are comparably sized (~10–15 M), so the improvement is not from scale.

TABLE II
SIMSHIFT HEATSINK, MEDIUM (OOD) SPLIT. BASELINES INCLUDE UDA; OURS DOES NOT. LOWER IS BETTER.

Model	UDA	Temp RMSE (K)	Vel. RMSE (m/s)
SurfaceGeoTransolver	✗	**12.14**	0.044
UPT (prev. best) [4]	✓	12.41	0.039
Transolver [3]	✓	13.43	0.041
PointNet [1]	✓	17.43	0.044

The benchmark number is itself out of distribution. Table III separates in-distribution accuracy (4.29 K) from the shifted target (12.14 K zero-shot on unseen fin counts).

TABLE III
SOURCE (IN-DISTRIBUTION) VS. TARGET (OOD) DETAIL FOR OUR MODEL

Domain	*T* (K)	Vel.	p_{rgh}	mean nRMSE
source (in-dist.)	4.29	0.025	203	0.287
target (OOD)	12.14	0.044	2303	0.671

B. *One Model, Two Device Classes*

A single set of weights, distinguished only by the device flag and fluid/BC conditioning, spans a wall-temperature, buoyancy-driven heatsink in air and a heat-flux, forced-liquid cold plate. Evaluated per class on held-out cases from our corpus (there is no public cold-plate CHT benchmark), the model reaches the errors in Table IV. The cold-plate mean nRMSE (0.028) is close to the in-distribution value. These numbers are on our own corpus and are not comparable to the SIMSHIFT numbers.

TABLE IV
MULTI-GEOMETRY MODEL, HELD OUT PER DEVICE CLASS (OUR CORPUS)

Held-out class	T RMSE (K)	T nRMSE	mean nRMSE
cold plates	3.11	0.039	0.028
heatsinks	5.13	0.065	0.145
in-distribution (both)	1.72	0.022	0.027

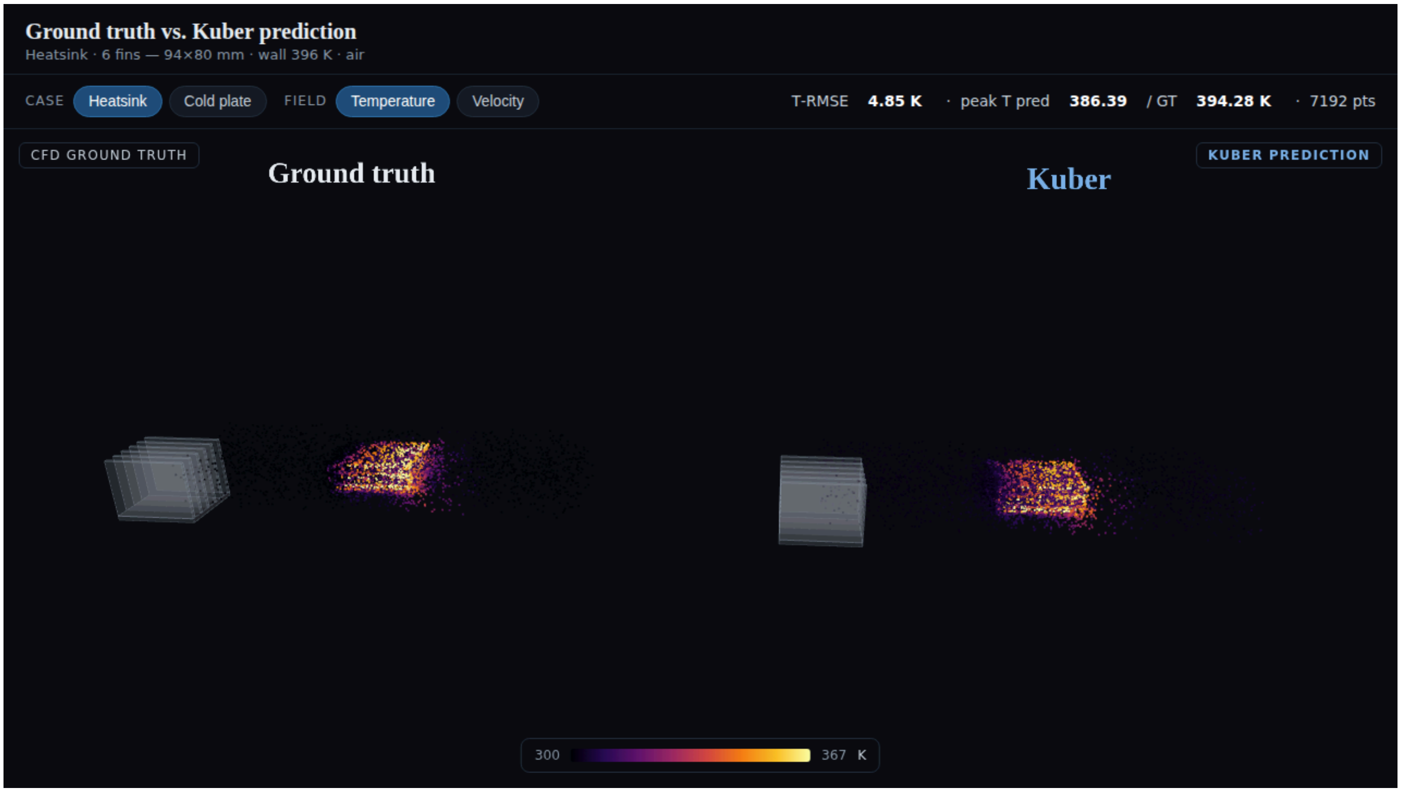

Fig. 3. Qualitative comparison from the interactive viewer: solid device geometry with the fluid temperature field, CFD ground truth (left) beside the surrogate prediction (right) for a heatsink; peak temperature and per-case T-RMSE are reported live.

C. Value of the Self-Generated Corpus

Does pretraining on the corpus help beyond the benchmark's own training set? A controlled A/B with the same model and SIMSHIFT fine-tuning, changing only the initialization (Table V). Pretraining helps materially at the easy shift (−19%) and in-distribution (−12%), modestly at medium, and is neutral at the hardest shift, which the corpus does not yet cover. The only changed ingredient is the corpus.

TABLE V
TEMPERATURE RMSE (K); FROM SCRATCH VS. CORPUS-PRETRAINED THEN FINE-TUNED

Shift	from scratch	pretrained	Δ
easy	8.99	**7.28**	−19%
medium	12.94	**12.38**	−4.3%
hard	14.42	14.43	±0
in-dist.	4.63	**4.09**	−12%

D. Numerical Stability

A deployable surrogate must reproduce steep near-wall gradients without over-smoothing or emitting unphysical spikes. We measure the ratio of predicted to CFD temperature-gradient magnitude in the edge band and at the steepest peaks (Table VI). Every ratio is at or below unity, in and out of distribution — the surrogate never produces a steeper-than-physical gradient — with an explosion fraction of exactly zero and no NaN/Inf. It is not merely over-smoothing: the in-distribution edge ratio (0.97 ≈ 1.0) reproduces the true edge structure, while out of distribution it errs slightly conservative, the safe direction.

TABLE VI
PREDICTED/CFD TEMPERATURE-GRADIENT RATIO AT EDGES (1.0 = FAITHFUL)

Metric	in-dist.	OOD
edge-band ∇T ratio	0.969	0.746
steepest-peak (p99.9) ratio	0.938	0.725
explosion fraction	0	0
NaN / Inf count	0	0

E. Speed and Data Fidelity

Inference is a single forward pass whose cost is independent of geometry, unlike CFD whose cost grows with mesh size. Against measured OpenFOAM solve times (median 22 minutes, up to 117 for fine geometries), a sub-second forward pass is roughly three to four orders of magnitude faster; we report the inference figure as an estimate pending exact per-GPU timing. On fidelity, three prism boundary layers recover the near-wall hot spot to within 0.1 K of a fine mesh (378.8 vs. 378.9 K) at ≈2.7× fewer cells.

VII. DISCUSSION AND LIMITATIONS

We state the boundaries plainly. The corpus is air-dominated (tens of oil/glycol cases), so the model is strongest on air. On

SIMSHIFT the OOD axis is fin count alone; leave-one-fluid-out and leave-one-shape-out remain future work. Cold plates are currently straight-channel, so the multi-geometry result shows device-class generality rather than topology extrapolation. On the averaged-nRMSE selector our temperature lead is bought with slightly higher velocity and pressure error, reported transparently. Inference latency is an estimate. The multi-geometry numbers are on our own corpus and are not comparable to the public benchmark.

VIII. Conclusion

Kuber packages the full stack needed to build a CHT surrogate — a reproducible data engine, a geometry-general surface-conditioned transformer, and an honest evaluation harness — and demonstrates it on electronics cooling, reaching state of the art on a public benchmark without domain adaptation and spanning two device classes from one model. The same interfaces target the broader class of coupled fluid–heat problems. Calibrated uncertainty, CAD connectors, and agentic geometry optimization are natural next steps; code, data, and an interactive viewer are released to support them.

References

[1] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE CVPR*, 2017.

[2] P. Setinek *et al.*, "SIMSHIFT: A benchmark for distribution shift in engineering simulation," preprint, 2025.

[3] H. Wu, H. Luo, H. Wang, J. Wang, and M. Long, "Transolver: A fast transformer solver for PDEs on general geometries," in *Proc. ICML*, 2024.

[4] B. Alkin, A. Fürst, S. Schmid, L. Gruber, M. Holzleitner, and J. Brandstetter, "Universal physics transformers," in *Proc. NeurIPS*, 2024.

[5] B. Alkin *et al.*, "Anchored/branched universal physics transformers," preprint, 2025.

[6] Z. Li *et al.*, "Fourier neural operator for parametric partial differential equations," in *Proc. ICLR*, 2021.

[7] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis, "Learning nonlinear operators via DeepONet," *Nature Machine Intelligence*, 2021.

[8] Z. Li *et al.*, "Geometry-informed neural operator for large-scale 3D PDEs," in *Proc. NeurIPS*, 2023.

[9] P. Lippe, B. Veeling, P. Perdikaris, R. Turner, and J. Brandstetter, "PDE-Refiner: Achieving accurate long rollouts with neural PDE solvers," in *Proc. NeurIPS*, 2023.

[10] NVIDIA, "PhysicsNeMo: A framework for physics-ML (GeoTransolver)," software, Apache-2.0.

[11] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.

[12] H. G. Weller, G. Tabor, H. Jasak, and C. Fureby, "A tensorial approach to computational continuum mechanics using object-oriented techniques," *Computers in Physics*, vol. 12, no. 6, 1998.